

SAPPHIRE AUTONOMY AND PERFORMANCE BENCHMARKS

Performance data for Sapphire ml.distributed across GPU clusters, model sizes, and precision modes.

1. Distributed Training Benchmarks

Model	Cluster	Strategy	MFU	Tok/sec	10T ETA
70B Dense	512x H100-80GB	TP1 PP1 DP512 ZeRO3 FP8	61.0%	~1.52M	~76 days
70B Dense	256x H100-80GB	TP1 PP1 DP256 ZeRO3 FP8	59.5%	~760K	~152 days
7B Dense	64x H100-80GB	TP1 PP1 DP64 ZeRO2 BF16	58.2%	~820K	~141 days
1T MoE	512x H100-80GB	TP4 PP8 DP16 EP64 ZeRO1 FP8	59.3%	~980K	~118 days
405B Dense	512x H200-141GB	TP8 PP4 DP16 ZeRO2 FP8	63.1%	~1.12M	~103 days
540B Dense	512x B200-192GB	TP8 PP8 DP8 ZeRO2 FP8	67.4%	~1.38M	~84 days
7B Dense	8x RTX4090-24GB	TP1 PP2 DP4 ZeRO3 BF16	41.7%	~280K	~413 days

MFU = Model FLOPs Utilization (% of theoretical peak FLOP/s). FP8 = 8-bit float via NVIDIA TransformerEngine.

2. Hardware Database

GPU	VRAM	TF32 TFLOP/s	BF16 TFLOP/s	FP8 TFLOP/s	NVLink GBps
H100-SXM5-80GB	80 GB	989	1979	3958	900
H200-SXM5-141GB	141 GB	989	1979	3958	900
B200-SXM6-192GB	192 GB	2250	4500	9000	1800
A100-SXM4-80GB	80 GB	312	624	N/A	600
RTX4090-24GB	24 GB	165	330	N/A	N/A

3. 70B Memory Breakdown (TP1 PP1 DP512 ZeRO-3 FP8, 512x H100-80GB)

Component	Per-GPU (GB)	Notes
Model Weights (FP8, ZeRO-3 sharded)	0.12 GB	140B bytes / 512 GPUs
Gradients (FP8, ZeRO-3 sharded)	0.12 GB	Sharded across DP ranks
Optimizer States (BF16, ZeRO-3)	1.45 GB	Adam: 2x model size / 512
Activations (seq 8192, batch 4)	35.0 GB	Main memory consumer
TOTAL	36.5 GB / 80 GB	54% headroom remaining

4. Precision Modes Comparison

Precision	FLOP/s (H100)	Memory (70B)	Stability	Notes
FP32	~495 TFLOP/s	~560 GB	Highest	Not practical for 70B+
BF16	~1979 TFLOP/s	~140 GB	Good	Standard for large models
FP8 (E4M3)	~3958 TFLOP/s	~70 GB	Good w/ scaling	2x BF16 throughput
INT8	~3958 TOPS	~70 GB	Inference only	Not for training